

# INNOVEX COMMAND CENTER

Product Requirements Document (PRD) — Version 2.4.0 (Enterprise Edition)

**Document Type:** Product Requirements Document (PRD)

**Status:** Approved for Production

**Date:** October 2026

## 1. Product Vision & Target Business Outcomes

**Innovex Command Center** is an enterprise-grade, closed-loop **AI-Powered Industrial Operations Optimization & Maintenance Intelligence Platform**. It bridges raw Internet of Things (IoT) sensor telemetry with executive business decision-making.

Unlike conventional predictive maintenance dashboards that stop at basic graph visualization and raw alert generation, Innovex executes a complete **closed-loop decision workflow**: detecting multi-sensor anomalies, forecasting Remaining Useful Life (RUL), delivering structured Explainable AI (XAI) root-cause diagnoses, cross-checking spare part lead times against RUL windows, modeling projected financial downtime risks, and automating work order creation.

- Zero Unplanned Downtime:** Eliminate catastrophic machine failures through early multi-sensor anomaly detection and dual-mode RUL forecasting.
- Stockout Mitigation:** Align spare part purchasing lead times with asset degradation curves to ensure zero maintenance delays due to missing components.
- Financial Risk Transparency:** Provide real-time financial impact projections (\$/hour downtime risk, SLA penalties, lost throughput) for operational delay scenarios.
- Operational Efficiency:** Reduce Mean Time To Repair (MTTR) by delivering actionable, prescriptive maintenance guidance directly to field technicians.

## 2. Target User Personas & User Journeys

| Persona                           | Core Responsibilities                                                           | Key Platform Needs                                                                                                 | Primary Pain Points Solved                                                                        |
|-----------------------------------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Plant Operations Director         | Overall manufacturing productivity, OEE targets, site financial P&L.            | High-level Operational Health KPIs, plant downtime financial projections, site risk rankings.                      | Lack of financial context for technical sensor alerts; inability to quantify downtime risk.       |
| Maintenance Engineer              | Asset health monitoring, diagnostics, RUL tracking, root-cause troubleshooting. | Feature-engineered sensor trends, hybrid anomaly scores, structured XAI root causes, RUL metrics.                  | Alert fatigue from raw threshold breaches; lack of diagnostic transparency in black-box ML tools. |
| Maintenance Supervisor            | Work order scheduling, technician assignment, calendar management.              | Interactive Maintenance Calendar, work order automation, priority-based dispatching.                               | Disconnect between ML anomaly alerts and CMMS/ERP work order scheduling systems.                  |
| Supply Chain / Inventory Manager  | Spare parts inventory management, reorder points, supplier lead times.          | Spare part readiness indicators ( <b>READY</b> , <b>AT RISK</b> , <b>UNAVAILABLE</b> ), lead time vs RUL matching. | Emergency stockouts occurring when lead time exceeds remaining operating hours.                   |
| Data Quality / Operations Analyst | Sensor data integrity monitoring, dataset uploads, anomaly validation.          | Data Quality Portal, completeness/outlier scoring, automated schema detection.                                     | Corrupted or noisy sensor streams triggering false positive anomaly alerts.                       |

### 3. System Architecture & Technology Stack

**Frontend Tier**

**React 18 + Vite + Tailwind CSS**

- Responsive Slate/Zinc industrial dark mode theme
- Recharts interactive time-series visualizations
- Lucide React iconography
- Sub-second UI page state transitions

**Node.js Express API Gateway**

**Node.js + Express + Multer**

- RESTful API routes & JWT authentication
- Role-based access control (RBAC)
- Audit logging engine for operational events

**Python FastAPI AI Engine**

**Python 3.11 + FastAPI + Scikit-Learn**

- 42 engineered telemetry features
- Scikit-Learn Isolation Forest & HistGradientBoosting
- Z-score multivariate distance metric

**Dual Storage Engine**

**MongoDB Atlas + In-Memory Fallback**

- MongoDB Atlas cloud cluster (Primary)
- Enterprise In-Memory Data Store (Zero-downtime fallback)

### 4. The Closed-Loop Operational Workflow

**End-to-End Decision Loop:**

SENSOR TELEMETRY → DATA QUALITY EVALUATION → FEATURE ENGINEERING → HYBRID ANOMALY DETECTION → FAILURE RISK SCORING → DUAL-MODE RUL FORECAST → STRUCTURED XAI DIAGNOSIS → PRESCRIPTIVE ACTION → SPARE PART READINESS → WORK ORDER CREATION → FINANCIAL RISK DELTA

### 5. Functional Requirements (FRs)

#### 5.1 FR-1: Dataset Management & Data Quality Portal

- **FR-1.1:** Support file uploads in `.csv`, `.xlsx`, and `.xls` formats up to 25MB.
- **FR-1.2:** Automated schema detection routing data to modules ( `inventory`, `maintenance`, `logistics`, `production`, `reflow` ).
- **FR-1.3:** 0–100 Data Quality Score based on completeness, timestamp continuity, and outlier distribution.
- **FR-1.4:** Persistent Delete Dataset capability with confirmation modal, unlinking stored files, purging child records, and toast notifications.

#### 5.2 FR-2: Asset Digital Twin & Telemetry Stream

- **FR-2.1:** Render operational Digital Twins for industrial assets (Compressors, Presses, CNC Mills, Robotic Arms).
- **FR-2.2:** Display telemetry parameters: Vibration RMS, Thermal Gradient, Line Pressure, Current Variance, Spindle RPM, Load.
- **FR-2.3:** Highlight parameters breaching baseline operational envelopes ( $> \pm 2.0\sigma$ ).

#### 5.3 FR-3: Hybrid AI Anomaly Engine & Dual-Mode RUL

- **FR-3.1:** Compute composite anomaly score (0–100) using weighted ensemble:  
`Composite Score = 0.45 * Model A (Z-Score) + 0.35 * Model B (Isolation Forest) + 0.20 * Model C (Trend)`
- **FR-3.2:** 42 feature extractions: 3-step rolling means, rolling std dev, RMS ( $\sqrt{\text{mean}(x^2)}$ ), sensor gradients ( $\Delta \text{hr}$ ), and correlation.
- **FR-3.3:** Dual-Mode RUL: `HistGradientBoostingRegressor` (Primary,  $\geq 20$  samples) vs Transparent Physical Degradation Rate (Fallback,  $< 20$  samples).

5.4 FR-4: Structured Explainable AI (XAI) & Prescriptive Actions

- **FR-4.1:** Non-black-box root causes: What Happened, Why Happened (breached sensor metrics), Failure Probability (%), Risk If Delayed.
- **FR-4.2:** Step-by-step prescriptive maintenance instructions.

5.5 FR-5: Spare Parts Inventory Matching & Readiness

- **FR-5.1:** Match equipment components to live spare parts inventory records.
- **FR-5.2:** Evaluate Readiness Status: READY (lead time  $\leq$  RUL), AT RISK (lead time approaches RUL), UNAVAILABLE (lead time > RUL).

5.6 FR-6: Maintenance Scheduling & Work Orders

- **FR-6.1:** 1-click work order creation directly from AI diagnostic cards into the Maintenance Calendar.
- **FR-6.2:** Interactive calendar with filtering by priority, status, and technician.
- **FR-6.3:** Dynamic overdue calculation based on system time vs scheduled date.

5.7 FR-7: What-If Financial Risk & Scenario Simulator

- **FR-7.1:** Simulate operational decisions (maintenance delay hours, procurement price surges, production adjustments).
- **FR-7.2:** Compare Baseline vs Simulated metrics across downtime cost, production loss, SLA penalties, and repair expense.
- **FR-7.3:** Display Financial Risk Delta (e.g., "\$18,400 Financial Risk Delta if delayed 24h").

6. Non-Functional Requirements (NFRs)

- **Latency:** Analytical execution  $\leq 350\text{ms}$ ; API response  $\leq 50\text{ms}$ ; UI render  $\leq 100\text{ms}$ .
- **High Availability:** Automatic zero-downtime failover to Enterprise In-Memory Data Store upon MongoDB cloud disconnection.
- **Security:** JWT token authentication, Role-Based Access Control (RBAC), audit logging engine.

7. Model Performance Benchmarks

| Metric                    | Target Benchmark      | Measured Performance | Status |
|---------------------------|-----------------------|----------------------|--------|
| Anomaly Precision         | $\geq 90\%$           | 92.0%                | PASS   |
| Anomaly Recall            | $\geq 85\%$           | 88.0%                | PASS   |
| F1 Score                  | $\geq 0.88$           | 0.90                 | PASS   |
| False Positive Rate (FPR) | $\leq 5.0\%$          | 4.0%                 | PASS   |
| RUL MAE                   | $\leq 4.0\text{ hrs}$ | 2.4 hrs              | PASS   |
| RUL RMSE                  | $\leq 5.0\text{ hrs}$ | 3.1 hrs              | PASS   |
| R <sup>2</sup> Score      | $\geq 0.85$           | 0.89                 | PASS   |

8. Deployment & Execution Commands

| Service Tier            | Port | Execution Command                                                                                                  |
|-------------------------|------|--------------------------------------------------------------------------------------------------------------------|
| FastAPI AI Microservice | 8000 | <code>cd ai-service &amp;&amp; .\.venv\Scripts\activate &amp;&amp; uvicorn app.main:app --reload -port 8000</code> |

| Service Tier                   | Port | Execution Command                             |
|--------------------------------|------|-----------------------------------------------|
| <b>Node.js Express Backend</b> | 5000 | <code>cd server &amp;&amp; npm run dev</code> |
| <b>React + Vite Client</b>     | 5173 | <code>cd client &amp;&amp; npm run dev</code> |